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NestedVLA: Learning to Consolidate and Generate Skills for Vision-Language-Action Models

A Nested Hypernetwork Generates Task-Adaptive Parameters Post Hoc, Outperforming Model Merging by 41 Percentage Points

By Guodong Du & Wanyu Lin (WanyuGroup)

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Teaching a robot to consolidate skills learned across heterogeneous tasks has long been limited by cross-platform data splits, gradient interference in joint training, and parameter conflicts in model merging. NestedVLA sidesteps all three by learning skills independently and consolidating post hoc through *generation* rather than aggregation: a nested hypernetwork, conditioned on task context, synthesises task-adaptive action-expert parameter deltas on demand and composes them with a frozen base VLA. On MetaWorld, ManiSkill, and CALVIN combined, NestedVLA outperforms the best model-merging baseline by an average of 41 percentage points. Accepted at NeurIPS 2026.

AT A GLANCE

Problem: Consolidating VLA skills across heterogeneous tasks causes gradient interference in joint training and parameter conflicts in model merging.

Why it matters: A general-purpose robot must master dozens of skills without one degrading another; current merging methods degrade all.

Method: Learn skills independently; train a nested hypernetwork to generate task-specific parameter deltas on demand; compose with frozen base VLA.

Result: +41 pp average success rate over strongest model-merging baseline across MetaWorld, ManiSkill, and CALVIN.

Evidence: Ablations confirm generation over aggregation is the key factor; adding new skills post hoc requires no global retraining.

The Big Picture

A generalist robot manipulation agent must be a jack of all trades: grasping fragile objects, pushing heavy boxes, drawing precise trajectories, stacking irregular items. Each skill is typically learned from a dataset collected on specific hardware, with a specific camera setup, and sometimes on a completely different robot platform.

The dream is a single model that holds all these skills simultaneously. The nightmare is that every approach to combining them introduces a new failure mode: joint training causes gradient interference; model merging causes parameter conflicts; mixture-of-experts routing requires end-to-end co-training of all experts from the start.

NestedVLA's key insight is to separate skill *learning* from skill *consolidation*, and to implement consolidation as parameter generation rather than parameter aggregation.

Why Existing Approaches Fall Short

Joint multi-task training: Gradient interference arises when different skills require conflicting parameter updates; catastrophic forgetting prevents stable incorporation of new skills after initial training.

Model merging (TIES, DARE, task arithmetic): These methods produce a single merged checkpoint by combining parameter deltas; when deltas conflict in sign and magnitude, the merged parameters satisfy no task well.

Mixture of Experts with routing: All experts must be trained end-to-end simultaneously; post-hoc addition of a new skill requires full retraining to update the router.

Adapter methods: Per-task adapters do not share structure; the number of parameters grows linearly with the number of tasks.

Core Mechanism

Stage 1 — Independent skill learning: Each skill k is trained independently on its own data $\mathcal{D}_k$, producing a parameter delta $\Delta\theta_k^*$ relative to a shared base VLA θ_0.

Stage 2 — Hypernetwork-based consolidation: A nested hypernetwork $\mathcal{H}_\phi$ is trained to reconstruct any $\Delta\theta_k^*$ from task context τ_k and a compressed skill token $e_k = \text{Enc}(\Delta\theta_k^*)$. The skill tokens are stored in a compact library. At inference, the hypernetwork generates the appropriate delta for the queried task and composes it additively with the frozen base:

$$\theta(\tau) = \theta_0 + \mathcal{H}_\phi(\tau, \{e_k\})$$

VLM residuals: Task-specific shifts in text-conditioned attention are stored separately as compressed vectors alongside the skill tokens, capturing linguistic context the action expert alone cannot reconstruct.

Technical Formulation

For skill k, the effective policy parameters are $\theta_k = \theta_0 + \Delta\theta_k^*$, and the compressed skill token is $e_k = \text{Enc}(\Delta\theta_k^*)$.

The hypernetwork $\mathcal{H}_\phi$ is trained with a combined objective:

$$\mathcal{L}(\phi) = \mathbb{E}_{k,(s,a)\sim\mathcal{D}_k}\big[\|\hat{\Delta}\theta(\tau_k) - \Delta\theta_k^*\|_F^2 + \lambda_{\text{BC}}\,\mathcal{L}_{\text{BC}}(\theta_0 + \hat{\Delta}\theta(\tau_k);\, s, a)\big]$$

where $\hat{\Delta}\theta(\tau_k) = \mathcal{H}_\phi(\tau_k, \{e_k\})$, $\|\cdot\|_F$ is the Frobenius norm of the delta, and $\mathcal{L}_{\text{BC}}$ is the behavioural cloning loss on skill-k demonstrations.

The delta compression uses a patch tokenizer: $\Delta\theta_k \in \mathbb{R}^{P\times Q} \mapsto e_k \in \mathbb{R}^d$, where $d \ll PQ$. Reconstruction quality vs. storage is governed by the compression ratio $r = d/(PQ)$.

Learning or Inference Procedure

1. Train each skill independently on $\mathcal{D}_k$ to obtain $\Delta\theta_k^*$.
2. Compress each delta: $e_k = \text{Enc}(\Delta\theta_k^*)$.
3. Train $\mathcal{H}_\phi$ on all $(e_k, \tau_k, \Delta\theta_k^*)$ triples.
4. At inference: encode task context τ; retrieve relevant e_k; run $\mathcal{H}_\phi$ to obtain $\hat{\Delta}\theta$; execute with $\theta_0 + \hat{\Delta}\theta$.
5. To add a new skill: train it independently, encode its delta, add token to the library. No global retraining required.

What the Guarantee Says

The Frobenius norm objective provides a formal recovery guarantee: if $\mathcal{H}_\phi$ achieves zero training loss, then at test time it recovers $\Delta\theta_k^*$ exactly for any task whose context τ_k was seen during training. In practice, the reconstruction error is a measure of how well the hypernetwork generalises to novel task phrasings; the paper shows generalisation holds empirically to held-out phrasings not seen in training.

Experimental Findings

- **MetaWorld (50 tasks):** NestedVLA outperforms task arithmetic, TIES, and DARE; gains are largest on dexterous tasks where merging creates severe delta conflicts.
- **ManiSkill:** Consistent improvements across diverse manipulation categories; the hypernetwork generalises to task phrasings not seen during consolidation training.
- **CALVIN:** Language-conditioned manipulation; VLM residuals contribute strongly here, confirming that the linguistic context they carry is not recoverable from the action-expert delta alone.
- **Average gain:** +41 pp over the strongest merging baseline across all three benchmarks combined.

Ablations and Interpretation

Without hypernetwork (direct merge): Returns to merging baseline performance; the generation step is the critical ingredient.

Without independent pretraining (jointly trained): Hypernetwork training becomes unstable; the fixed deltas serve as stable anchors for the generation objective.

Without VLM residuals: Performance drops on CALVIN; the action expert alone cannot reconstruct task-specific language-conditioned attention.

Compression ratio sweep: Aggressive compression ($r < 0.05$) degrades fine-grained dexterous tasks while performing well on coarse manipulation; a ratio of $r \approx 0.1$ is the practical sweet spot.

Why generation beats aggregation: Parameter averaging smooths task-specific directions in parameter space; a hypernetwork preserves them by mapping context to the appropriate structured delta directly.

Reference

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